

DTSA 5719

Computing, Ethics, and Society 2 - Algorithmic Bias and Professional Ethics

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (51.2 / 100)	46.1 / 100
SPECIALIZATION Computing, Ethics, and Society		

46

CURRENT AVG SCORE
across 6 job roles

81

POTENTIAL SCORE
with recommended additions

+40

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Computing systems and technologies fundamentally impact the lives of most people in the world, including how we communicate, get information, socialize, and receive healthcare. This course is the second of a three course sequence that examines ethical issues in the design and implementation of computing systems and technologies, and reflects upon the broad implication of computing on our society. It covers algorithmic bias in machine learning methods, professional ethics, and issues in the tech workplace.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	51.2	85.4	+34.1	HIGH
Machine Learning Engineer	49.4	75.6	+26.2	MEDIUM
AI Engineer	49.1	86.9	+37.7	HIGH
MLOps Engineer	46.7	86.6	+40.0	HIGH
Data Scientist	45.1	82.6	+37.5	HIGH
Applied Scientist	35.3	67.5	+32.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

ML Engineer	51.2 → 85.4 (+34.1 pts)
media data model engineering infrastructure problems product technical	
Machine Learning Engineer	49.4 → 75.6 (+26.2 pts)
data web engineering tools science model applications technical	
AI Engineer	49.1 → 86.9 (+37.7 pts)
data robotics engineering infrastructure software technical model problems	
MLOps Engineer	46.7 → 86.6 (+40.0 pts)
infrastructure data model media engineering tools	
Data Scientist	45.1 → 82.6 (+37.5 pts)
data product different social engineering problems approaches	
Applied Scientist	35.3 → 67.5 (+32.2 pts)
computational problems technical data engineering regulatory studies science	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.